

- (b) $^{35}_{17}\text{Cl}$ and $^{36}_{17}\text{Cl}$
- (c) $^{20}_{10}\text{Ne}$ and $^{17}_{10}\text{Ne}$
- (d) $^{40}_{20}\text{Ca}$ and $^{45}_{20}\text{Ca}$
- (e) $^{195}_{80}\text{Hg}$ and $^{196}_{80}\text{Hg}$

Answer

- (a) The $^{16}_8\text{O}$ contains 8 protons and 8 neutrons (even-even) and the $^{17}_8\text{O}$ contains 8 protons and 9 neutrons (even-odd). Therefore, $^{17}_8\text{O}$ is radioactive.
- (b) The $^{35}_{17}\text{Cl}$ has 17 protons and 18 neutrons (odd-even) and the $^{36}_{17}\text{Cl}$ has 17 protons and 19 neutrons (odd-odd). Hence, $^{36}_{17}\text{Cl}$ is radioactive.
- (c) The $^{20}_{10}\text{Ne}$ contains 10 protons and 10 neutrons (even-even) and the $^{17}_{10}\text{Ne}$ contains 10 protons and 7 neutrons (even-odd). Therefore, $^{17}_{10}\text{Ne}$ is radioactive.
- (d) The $^{40}_{20}\text{Ca}$ has even-even situation and $^{45}_{20}\text{Ca}$ has even-odd situation. Thus, $^{45}_{20}\text{Ca}$ is radioactive.
- (e) The $^{195}_{80}\text{Hg}$ has even number of protons and odd number of neutrons and the $^{196}_{80}\text{Hg}$ has even number of protons and even number of neutrons. Therefore, $^{195}_{80}\text{Hg}$ is radioactive.

Nuclear Binding Energy

The nuclear binding energy is an energy required to break up a nucleus into its components protons and neutrons. In essence, it is a quantitative measure of the nuclear stability. The concept of nuclear binding energy is based on Einstein's famous equation, $E = mc^2$, where E is the energy, m is the mass and c is the velocity of light, and according to which the energy and mass are inter-convertible.

Nucleus contains mainly two particles – protons and neutrons- in addition to many other elementary particles. Thus, the mass of the nucleus is primarily comes from the masses of protons and neutrons. But the experiments have shown that the sum of the masses of protons and neutrons is always greater than experimentally determined nuclear mass. Why is it so? The answer to this question lies in the way the nature creates nucleus. When nature creates nucleus, it takes protons and neutrons and binds them together and puts them in a tiny space called nucleus. In order to bind protons and neutrons together, some energy is needed, which is taken out of the masses of protons and neutrons. It means that nature is very smart, it does not spend any of its own energy rather it converts some of the masses of protons and neutrons into an energy and utilizes that energy to